Juan Augusto Paredes Salazar

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Name variants: Juan Paredes; Juan Augusto Paredes; Juan A. Paredes; J. A. Paredes; J. A. Paredes Salazar; J.A.P. Salazar

Research Interests

Theory: Feedback Control Systems, Adaptive Control, Optimal Control, Model Predictive Control, Data-Driven Learning for Control, Control Barrier Functions for Safe Control Synthesis, Trajectory Optimization, System Identification, Machine Learning, Kalman Filtering, Nonlinear State Estimation.

Applications: Unmanned Aerial Vehicles, Multicopters, Fixed-wing Aircraft, Vertical Takeoff and Landing (VTOL) Aircraft, Robotic Systems, Vibration Suppression Systems, Self-Excited Systems

Education

2019 - 2023 **Ph.D., Aerospace Engineering (Adaptive Control for Aerospace Applications)**, ***University of Michigan***, Ann Arbor, Michigan, GPA: 4.0/4.0
Dissertation title: Adaptive Control of Self-Excited Systems with Application to a Gas Turbine Combustor

2017 - 2019 **M.S.E., Aerospace Engineering (Controls and Dynamics)**, ***University of Michigan***, Ann Arbor, Michigan, GPA: 4.0/4.0

2010 - 2015 **B.S.E., Mechatronics Engineering**, ***Pontifical Catholic University of Peru***, Lima, Peru, GPA: 3.58/4.0

Teaching Experience

2026 **Adjunct Professor**, ***Course: ENME 408 - UAV Dynamics & Control***, University of Maryland, Baltimore County, Spring 2026
This hands-on course develops the foundational skills needed to model, analyze, simulate, and fly multicopters.

2025 **Adjunct Professor**, ***Course: ENME 221 - Dynamics***, University of Maryland, Baltimore County, Spring 2025
ENME 221 is an undergraduate-level course introducing dynamics of particles and rigid bodies. This course lays the theoretical foundation to derive the equations of motion of mechanical systems using a coordinate-free approach.

2020 - 2022 **Graduate Student Instructor**, University of Michigan
Courses:
- AEROSP 540: Intermediate Dynamics (Fall 2021, Fall 2022)
- AEROSP 584: Navigation and Guidance of Aerospace Vehicles (Fall 2020)

2015 - 2017 **Teacher's Assistant**, Pontifical Catholic University of Peru
Courses:
- Autonomous Control
- Computer Integrated Manufacturing
- Introduction to Aeronautical Engineering

Employment History

Since 2026 **Postdoctoral Research Fellow**, *Research Laboratory led by Professor Dennis S. Bernstein,*
University of Michigan, Ann Arbor, Michigan

2024 - 2026 **Post-doctoral Research Associate**, *Research Laboratory led by Professor Ankit Goel,*
University of Maryland, Baltimore County, Baltimore, Maryland

2023 - 2024 **Postdoctoral Research Fellow**, *Research Laboratory led by Professor Dennis S. Bernstein,* *University of Michigan*, Ann Arbor, Michigan

2019 - 2023 **Ph.D. Candidate**, *Research Laboratory led by Professor Dennis S. Bernstein,* *University of Michigan*, Ann Arbor, Michigan

2016 - 2017 **Laboratory Assistant**, *Unmanned Aerial Systems Laboratory,* *Pontifical Catholic University of Peru*, Lima, Peru

2015 - 2017 **Project Assistant**, *Biomechanics and Applied Robotics Laboratory,* *Pontifical Catholic University of Peru*, Lima, Peru

Patent Applications

2026 A. Goel and J. A. Paredes Salazar, "Real-Time In-Situ Vibration Suppression System," U.S. Provisional Patent Application No. 64/094,677, filed June 19, 2026. Applicant: University of Maryland, Baltimore County.

Publications

Journal Manuscripts Under Review

[J1] J. A. Paredes Salazar and A. Goel, "Performance-based adaptation termination for preventing parameter drift in adaptive vibration suppression," *arXiv preprint arXiv:2608.02570*, submitted to J. Sound Vib.

[J2] J. Usevitch, J. A. Paredes Salazar, and A. Goel, "Computing safe control inputs using discrete-time matrix control barrier functions via convex optimization," *arXiv preprint arXiv:2510.09925*, submitted to IEEE Trans. Autom. Control.

Journal Publications

[J3] J. A. Paredes, O. Kouba, and D. S. Bernstein, "Self-excited dynamics of discrete-time Lur'e systems with affinely constrained, piecewise-C1 feedback nonlinearities," *IEEE Open J. Contr. Sys.*, vol. 3, pp. 214–224, 2024.

[J4] J. A. Paredes, R. Ramesh, M. Gamba, and D. S. Bernstein, "Experimental application of a quasi-static adaptive controller to a Dual Independent Swirl combustor," *Comb. Sci. Tech.*, vol. 197, no. 9, pp. 2116–2149, 2024.

[J5] J. A. Paredes, Y. Yang, and D. S. Bernstein, "Output-only identification of self-excited systems using discrete-time Lur'e models with application to a gas-turbine combustor," *Int. J. Contr.*, vol. 97, no. 2, pp. 187–212, 2024.

[J6] J. A. Paredes and D. S. Bernstein, "Experimental implementation of retrospective cost adaptive control for suppressing thermoacoustic oscillations in a Rijke tube," *IEEE Trans. Contr. Sys. Tech*, vol. 31, no. 6, pp. 2484–2498, 2023, DOI: 10.1109/TCST.2023.3262223.

[J7] J. A. Paredes, P. Sharma, B. Ha, M. Lanchares, E. Atkins, P. Gaskell, and I. Kolmanovsky, "Development, implementation, and experimental outdoor evaluation of quadcopter controllers for computationally limited embedded systems," *Annu. Rev. Contr.*, vol. 52, pp. 372–389, 2021.

Accepted Conference Papers

[C1] C. Carrillo, M. Koca, O. Tumuklu, O. Moran, J. A. Paredes Salazar, and A. Goel, "Dynamic mode adaptive control for pressure-recovery regulation in a supersonic diffuser," accepted for presentation at the 2027 AIAA SciTech Forum.

[C2] O. Moran, J. A. Paredes Salazar, and A. Goel, "Adaptive vibration suppression in soft-supported beam models of helicopter airframes," accepted for presentation at the 2027 AIAA SciTech Forum.

[C3] J. A. Paredes Salazar, O. Moran, and A. Goel, "Model-free vibration suppression in a cantilever beam using dynamic mode adaptive control," accepted for presentation at the 2027 AIAA SciTech Forum.

[C4] M. Moody, O. Moran, J. A. Paredes Salazar, and A. Goel, "Modeling and data-driven flutter suppression in a pitch–plunge aeroelastic system," accepted for presentation at the 2027 AIAA SciTech Forum.

Conference Publications

[C5] J. A. Paredes Salazar, J. Usevitch, and A. Goel, "Predictive control barrier functions for discrete-time linear systems with unmodeled delays," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2026, pp. 3876–3881.

[C6] S. Verma, J. A. P. Salazar, J. M. P. Delgado, A. Goel, and D. S. Bernstein, "Sensor-noise mitigation in extremum seeking control using adaptive numerical differentiation," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2026, pp. 3164–3169.

[C7] J. A. Paredes Salazar and A. Goel, "Model-free adaptive output feedback vibration suppression in a cantilever beam," in *AIAA SCITECH 2026 Forum*, 2026, p. 1845.

[C8] J. M. Portella Delgado, J. A. Paredes Salazar, and A. Goel, "Extremum seeking framework with vanishing dither signal," in *AIAA SCITECH 2026 Forum*, 2026, p. 1759.

[C9] M. Mirtaba, J. A. Paredes Salazar, D. Huang, and A. Goel, "Low-order $\mathcal{H}_2/\mathcal{H}_\infty$ controller design for aeroelastic vibration suppression," in *AIAA SCITECH 2026 Forum*, 2026, p. 1846.

[C10] J. A. Paredes Salazar, A. Goel, R. Costich, M. Koca, O. Tumuklu, and M. Amitay, "Data-driven pressure recovery regulation in diffusers," in *AIAA SCITECH 2026 Forum*, 2026, p. 1137.

[C11] A. Phelps, J. A. Paredes Salazar, and A. Goel, "Data-driven fuzzy control for time-optimal aggressive trajectory following," in *Proc. Model. Est. Contr. Conf. (MECC)*, IFAC-PapersOnLine, 2025, pp. 473–478.

[C12] M. Mirtaba, P. Oveissi, J. A. Paredes Salazar, and A. Goel, "Single-shot learning of multirotor controller gains: A data-driven approach with experimental validation," in *Proc. IEEE Conf. Contr. Tech. Appl. (CCTA)*, IEEE, 2025, pp. 946–951.

[C13] J. A. Paredes Salazar and D. S. Bernstein, "Experimental application of predictive cost adaptive control to thermoacoustic oscillations in a Rijke tube with unknown input delay," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2025, pp. 1864–1869.

[C14] J. A. Paredes Salazar and A. Goel, "MPC-guided, data-driven fuzzy controller synthesis," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2025, pp. 46–51.

[C15] R. A. Alhazmi, J. A. Paredes Salazar, S. A. U. Islam, and D. S. Bernstein, "Application of root-finding methods to iterative model predictive control of pseudo-linear systems," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2025, pp. 3385–3390.

[C16] J. A. Paredes Salazar and D. S. Bernstein, "Absolute-stability-based closed-loop stability analysis of adaptive model predictive control for self-excited Lur'e systems," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2025, pp. 2477–2482.

[C17] S. A. U. Islam, J. A. Paredes Salazar, and D. S. Bernstein, "Multirate model predictive control of inner-outer loops," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2025, pp. 3335–3340.

[C18] R. Richards, J. Paredes, and D. Bernstein, "Predictive cost adaptive control of fixed-wing aircraft without prior modeling," in *Proc. AIAA SciTech Forum*, 2025, p. 2081.

[C19] Y. Y. Chee, P. Oveissi, S. Shao, J. Lee, J. A. Paredes, D. S. Bernstein, and A. Goel, "A Hammerstein-Weiner modification of adaptive autopilot for parameter drift mitigation with experimental results," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2024, pp. 1556–1561.

[C20] R. J. Richards, Y. Yang, J. A. Paredes, and D. S. Bernstein, "Output-only identification of Lur'e systems with hysteretic feedback nonlinearities," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2024, pp. 2891–2896.

[C21] Y. Y. Chee, J. A. Paredes, and D. S. Bernstein, "Data-driven retrospective-cost-based adaptive digital PID control," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2024, pp. 5163–5168.

[C22] J. A. Paredes, J. M. Portella Delgado, D. S. Bernstein, and A. Goel, "Retrospective cost-based extremum seeking control with vanishing perturbation for online output minimization," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2024, pp. 2344–2349.

[C23] K. F. Aljanaideh, M. Al Janaideh, R. J. Richards, J. A. Paredes, and D. S. Bernstein, "Output-only identification of lur'e systems with prandtl-ishlinskii hysteresis nonlinearities," *IFAC-PapersOnLine*, vol. 58, no. 15, pp. 366–371, 2024.

[C24] Y. Y. Chee, P. Oveissi, J. Paredes, D. S. Bernstein, and A. Goel, "Performance comparison of adaptive autopilot architectures for multicopter stabilization and trajectory tracking," in *Proc. AIAA SciTech Forum*, 2024, p. 1391.

[C25] J. Lee, J. Spencer, S. Shao, J. A. Paredes, D. S. Bernstein, and A. Goel, "Experimental flight testing of a fault-tolerant adaptive autopilot for fixed-wing aircraft," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2023, pp. 2981–2986.

[C26] J. A. Paredes, S. A. U. Islam, and D. S. Bernstein, "Adaptive stabilization of thermoacoustic oscillations in a Rijke tube," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2022, pp. 28–33.

[C27] J. Spencer, J. Lee, J. A. Paredes, A. Goel, and D. Bernstein, "An adaptive PID autotuner for multicopters with experimental results," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, IEEE, 2022, pp. 7846–7853.

[C28] J. A. Paredes, R. Ramesh, S. Obidov, M. Gamba, and D. Bernstein, "Experimental investigation of adaptive feedback control on a Dual-Swirl-Stabilized gas turbine model combustor," in *Proc. AIAA SciTech Forum*, 2022, p. 2058.

[C29] A. Goel, J. A. Paredes, H. Dadhaniya, S. A. U. Islam, A. M. Salim, S. Ravela, and D. S. Bernstein, "Experimental implementation of an adaptive digital autopilot," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2021, pp. 3737–3742.

[C30] J. A. Paredes and D. S. Bernstein, "Identification of self-excited systems using discrete-time, time-delayed Lur'e models," in *Proc. Amer. Contr. Conf. (ACC)*, IEEE, 2021, pp. 3939–3944.

[C31] R. Ramesh, J. A. Paredes, D. Bernstein, and M. Gamba, "Design and characterization of the Dual Independent Swirl Combustor facility (DISCo)," in *Proc. AIAA Prop. Energy Forum*, 2021, p. 3479.

[C32] J. A. Paredes, S. A. U. Islam, and D. S. Bernstein, "A time-delayed Lur'e model with biased self-excited oscillations," in *Proc. Amer. Contr. Conf. (ACC)*, Denver, Jul. 2020, pp. 2699–2704.

[C33] J. Gonzalez, A. Chavez, J. A. Paredes, and C. Saito, "Obstacle detection and avoidance device for multirotor UAVs through interface with Pixhawk flight controller," in *Proc. Conf. Autom. Sci. Eng. (CASE)*, Aug. 2018, pp. 110–115.

[C34] M. Abarca, C. Saito, A. Angulo, J. A. Paredes, and F. Cuellar, "Design and development of an hexacopter for air quality monitoring at high altitudes," in *Proc. Conf. Autom. Sci. Eng. (CASE)*, IEEE, 2017, pp. 1457–1462.

[C35] J. A. Paredes, C. Saito, M. Abarca, and F. Cuellar, "Study of effects of high-altitude environments on multicopter and fixed-wing UAVs' energy consumption and flight time," in *Proc. Conf. Autom. Sci. Eng. (CASE)*, IEEE, 2017, pp. 1645–1650.

[C36] L. C. Caballero, C. Saito, R. B. Micheline, and J. A. Paredes, "On the design of an UAV-based store and forward transport network for wildlife inventory in the western Amazon rainforest," in *Proc. Int. Congr. Electron. Elect. Eng. Comput. (INTERCON)*, IEEE, 2017, pp. 1–4.

[C37] J. A. Paredes, J. González, C. Saito, and A. Flores, "Multispectral imaging system with UAV integration capabilities for crop analysis," in *Proc. Int. Symp. Geosci. Remote Sens. (GRSS-CHILE)*, IEEE, 2017, pp. 1–4.

[C38] D. A. Flores, C. Saito, J. A. Paredes, and F. Trujillano, "Multispectral imaging of crops in the Peruvian Highlands through a fixed-wing UAV system," in *Proc. Int. Conf. Mechatron. (ICM)*, IEEE, 2017, pp. 399–403.

[C39] D. A. Flores, C. Saito, J. A. Paredes, and F. Trujillano, "Aerial photography for 3D reconstruction in the Peruvian Highlands through a fixed-wing UAV system," in *Proc. Int. Conf. Mechatron. (ICM)*, IEEE, 2017, pp. 388–392.

[C40] J. A. Paredes, C. Jacinto, R. Ramirez, I. Vargas, and L. Trujillano, "Fuzzy-PD controller for behavior mixing and improved performance in quadcopter attitude control systems," in *Proc. Latin Amer. Conf. Comput. Intell. (LA-CCI)*, IEEE, 2016, pp. 1–6.

[C41] J. A. Paredes, C. Jacinto, R. Ramirez, I. Vargas, and L. Trujillano, "Simplified fuzzy-PD controller for behavior mixing and improved performance in quadcopter attitude control systems," in *Proc. Tech. Sci. Conf. Andean Council (ANDESCON)*, IEEE, 2016, pp. 1–4.

[C42] J. A. Paredes, J. Acevedo, H. Mogrovejo, J. Villalta, and R. Furukawa, "Quadcopter design for medicine transportation in the Peruvian Amazon Rainforest," in *Proc. Int. Congr. Electron. Elect. Eng. Comput. (INTERCON)*, IEEE, 2016, pp. 1–6.

Service

Faculty Mentor for Summer Research Opportunity Program (SROP) at University of Michigan (Summer 2024)

Journal Reviewer:
- Annual Reviews in Control (ARC)
- Control Engineering Practice
- IEEE Control Systems Magazine (CSM)
- Engineering Applications of Artificial Intelligence (EAAI)
- International Journal of Control (IJC)
- IEEE Transactions on Automatic Control (TACON)
- IEEE Transactions on Automation Science and Engineering (T-ASE)
- IEEE Transactions on Control Systems Technology (TCST)
- International Journal of Power Electronics and Drive Systems (IJPEDS)
- International Journal of Robust and Nonlinear Control (IJRNC)
- Nonlinear Dynamics
- PLOS One
- Scientific Data
- Systems Science & Control Engineering

Conference Reviewer:
- American Control Conference (ACC)
- Conference on Control Technology and Applications (CCTA)
- Conference on Decision and Control (CDC)
- European Control Conference (ECC)
- International Conference on Robotics & Automation (ICRA)
- Symposium on System Identification (SYSID)

Conference Session Chair:
- Adaptive Control, 2025 American Control Conference, Denver, CO
- Model Predictive Control, 2025 American Control Conference, Denver, CO

Certifications

FAA — Remote Pilot Certificate with a Small UAS Rating (14 C.F.R. Part 107); required recurrent training completed in 2026.

Professional Associations

IEEE, Member, ID Number: 93821833

Tau Beta Pi, Member

Grants

2017 Research Project Annual Contest (CAP 2017), internal grant awarded by the Pontifical Catholic University of Peru (PUCP) for the **Obstacle detection and avoidance device for multirotor UAVs through interface with Pixhawk flight controller** project.

Computer Skills

Languages	C, C++, C#, Python, MATLAB, Bash, Fortran 90, LaTeX
Software	PX4 and ArduPilot firmware for UAS, dSPACE Simulink Interface, CasADi for Optimization, ROS, Arduino, Autodesk Inventor, SolidWorks, Simulink Real-Time, Eagle, LTspice, AutoCAD Mechanical, LabVIEW
Operating Systems	Windows, Ubuntu, Red Hat, CentOS, Linux for Embedded Systems
Manufacturing Technologies	3D Printing, Laser Cutting, CNC Machining, PCB Milling